

# Aditya Kulkarni

aditya.kulkarni.emb@gmail.com | +91-9321098765 | github.com/adityakulkarni-emb | Nagpur, India

CGPA: 8.3 / 10

## PROFESSIONAL SUMMARY

Embedded systems engineer with expertise in C/C++, ARM Cortex, RTOS, and hardware debugging. Built 6 embedded projects including a real-time sensor fusion system. Bosch intern. Strong in firmware development and low-level programming.

## EDUCATION

**B.Tech — Electronics and Computer Science**

2021 – 2025

VNIT Nagpur, NIT

## TECHNICAL SKILLS

|                      |                                                                  |
|----------------------|------------------------------------------------------------------|
| Embedded             | C, C++, Assembly (ARM), FreeRTOS, Zephyr RTOS                    |
| Microcontrollers     | STM32, ESP32, Arduino, Raspberry Pi, nRF52840                    |
| Protocols            | UART, I2C, SPI, CAN, USB, BLE, MQTT                              |
| Tools                | Keil uVision, STM32CubeIDE, OpenOCD, GDB, JTAG/SWD, Oscilloscope |
| Languages (software) | Python, MATLAB                                                   |
| PCB                  | KiCad, LTSpice                                                   |

## WORK EXPERIENCE

**Embedded Software Intern — Bosch India**

Jun 2024 – Aug 2024

(Automotive Electronics), Bengaluru

- Developed firmware for AUTOSAR-compliant CAN driver on STM32F4 processor.
- Wrote and executed unit tests (CppUTest) achieving 90%+ code coverage.
- Debugged timing-critical ISR routines using oscilloscope and logic analyzer.

## PROJECTS

### Real-Time Sensor Fusion (STM32)

- IMU + GPS sensor fusion using Kalman filter on STM32F407. FreeRTOS multitasking architecture.
- 500Hz sensor update rate with <2ms end-to-end latency.

### Smart Home Automation Hub (ESP32)

- BLE mesh network connecting 8 nodes. MQTT broker for cloud integration. OTA firmware updates.

### Battery Management System (BMS)

- Li-ion cell voltage/temperature monitoring firmware. Overcurrent and over-temperature protection with hardware fault handling.

## CERTIFICATIONS

- ARM Cortex-M Programming (Udemy, 2023)
- FreeRTOS Real-Time Programming (Udemy, 2024)
- MATLAB for Engineers (MathWorks, 2023)